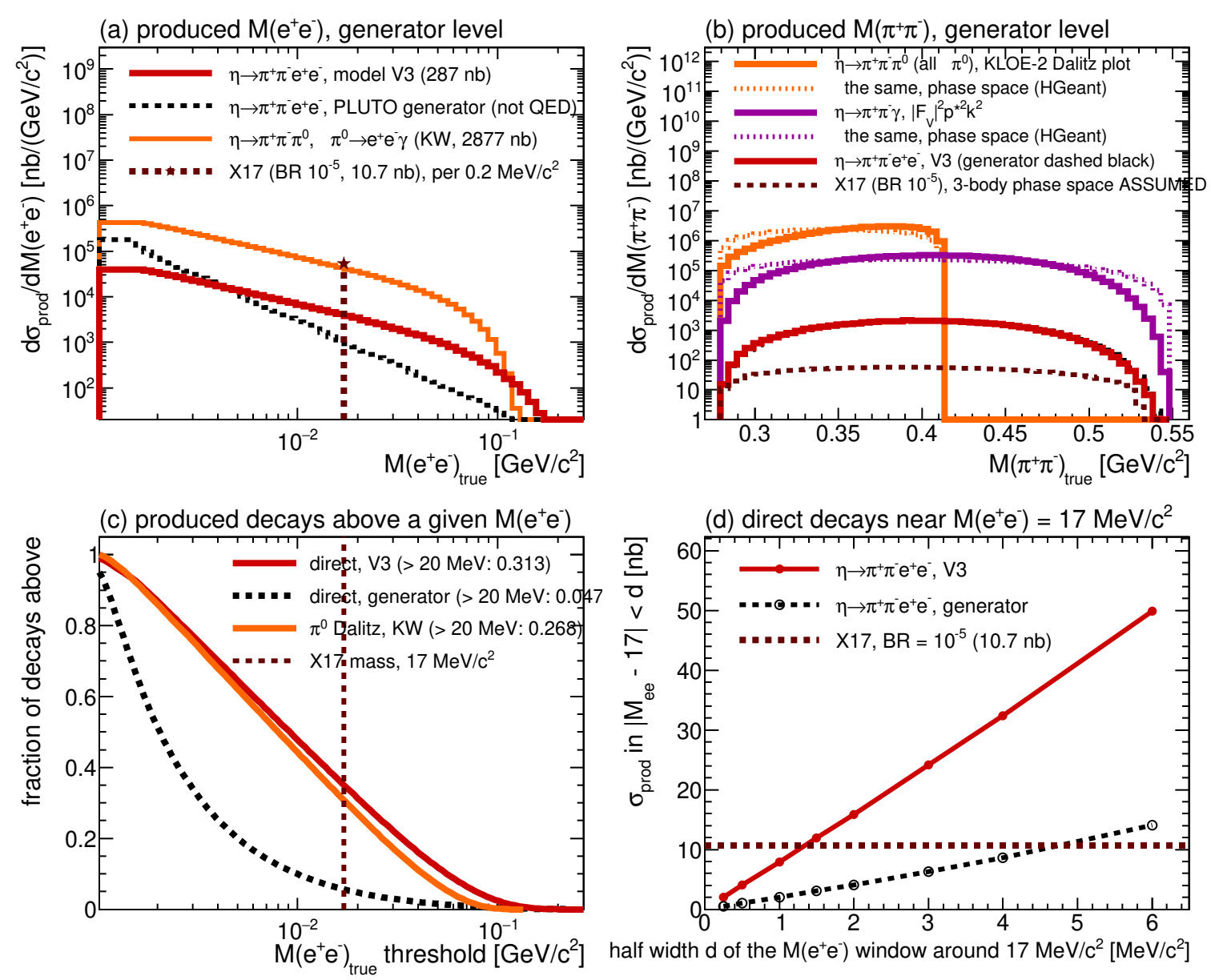
(a) produced M(e+e-), generator level
η→π+π-e+e-, model V3 (287 nb)
η→π+π-e+e-, PLUTO generator (not QED)
η→π+π-π0, π0→e+e-γ (KW, 2877 nb)
X17 (BR 10-5, 10.7 nb), per 0.2 MeV/c2
dσprod/dM(e+e-) [nb/(GeV/c2)]
M(e+e-)true [GeV/c2]
(b) produced M(π+π-), generator level
η→π+π-π0 (all π0), KLOE-2 Dalitz plot
the same, phase space (HGeant)
η→π+π-γ, |FV|2p*2k2
the same, phase space (HGeant)
η→π+π-e+e-, V3 (generator dashed black)
X17 (BR 10-5), 3-body phase space ASSUMED
dσprod/dM(π+π-) [nb/(GeV/c2)]
M(π+π-)true [GeV/c2]
(c) produced decays above a given M(e+e-)
direct, V3 (> 20 MeV: 0.313)
direct, generator (> 20 MeV: 0.047
π0 Dalitz, KW (> 20 MeV: 0.268)
X17 mass, 17 MeV/c2
fraction of decays above
M(e+e-)true threshold [GeV/c2]
(d) direct decays near M(e+e-) = 17 MeV/c2
η→π+π-e+e-, V3
η→π+π-e+e-, generator
X17, BR = 10-5 (10.7 nb)
σprod in |Mee - 17| < d [nb]
half width d of the M(e+e-) window around 17 MeV/c2 [MeV/c2]